

Vision-Only Latent Diffusion Super-Resolution without T2I Pretraining

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Completed-experiment snapshot through Stage 4 XL edge-loss training

Abstract

This report summarizes a work-in-progress x4 super-resolution system trained directly in vision space, without importing a pretrained text-to-image diffusion backbone. The completed pipeline consists of a factor-4 VAE, deterministic LR-to-HR-latent pretraining, conditional latent diffusion, and condition-start fine-tuning on a 103k-image photo split. The latest completed XL-stage result uses a 500M-class full inference path: a wider Stage 2 condition encoder trained with a stronger degradation curriculum and a deeper 469.6M-parameter diffusion U-Net fine-tuned with decoded edge/highpass losses. On the current `photo_v3_noise_mix` sampled validation setup, the Stage 4 XL edge-loss checkpoint improves over bicubic by +0.7195 dB PSNR.

1 Objective

The target task is single-image x4 restoration:

$$128 \times 128 \rightarrow 512 \times 512.$$

A later extension targets

$$192 \times 192 \rightarrow 768 \times 768.$$

The design goal is a practical latent diffusion SR model trained from project checkpoints only. The system intentionally avoids Stable Diffusion, T2I backbones, text conditioning, and imported generative priors from pretrained diffusion models.

2 Architecture

The model is split into three trainable parts:

$$\text{HR image} \rightarrow \text{VAE} \rightarrow \text{HR latent},$$

$$\text{LR image} \rightarrow \text{condition encoder} \rightarrow \text{condition latent},$$

$$\text{noisy HR latent, condition latent, } t, \text{ domain id} \rightarrow \text{conditional U-Net}.$$

The VAE decoder maps the denoised latent back to the 512px HR output. Domain IDs are implemented for photo and anime/illustration data, although the completed large-scale runs reported here use the photo split.

3 Training Stages

The project is organized as staged training rather than a single end-to-end run:

1. Stage 1 trains the factor-4 VAE on HR crops.

2. Stage 2 trains a deterministic condition encoder to predict HR latents from degraded LR inputs.
3. Stage 3 trains a conditional diffusion U-Net in latent space.
4. Stage 4 fine-tunes diffusion from the condition latent at low timesteps.
5. XL variants widen and deepen the condition encoder or diffusion U-Net while using partial checkpoint initialization where tensor shapes match.

4 Completed Implementation

The repository currently includes:

- manifest-based dataset loading with explicit domain IDs,
- on-the-fly x4 degradation pipelines,
- a factor-4 AutoencoderKL with 16 latent channels,
- deterministic LR-to-HR-latent pretraining,
- conditional latent diffusion with timestep and domain conditioning,
- DDIM/img2img sampling for validation,
- condition-start Stage 4 fine-tuning,
- bf16 CUDA/ROCm training paths,
- W&B logging, fixed validation samples, checkpoint evaluation, and recovery scripts for VM migration,
- partial checkpoint initialization for widened/deepened models,
- PyTorch DDP diffusion training with single-GPU fallback.

5 Data and Degradation

The large-scale photo runs use a 103,450-image training split and a fixed 100-image validation split. HR crops are 512px and LR inputs are 128px. The original photo degradation was extended with stronger blur, noise, compression, ringing, color shift, and banding variants. The latest XL condition encoder uses the `photo_v3_noise_mix` degradation preset to emphasize denoise/sharpening behavior.

Item	Value
Task	x4 SR, 128px LR to 512px HR
Active large-scale domain	Photo
Train split	103,450 images
Validation split	100 fixed images
VAE latent	128px, 16 channels
Primary dtype	bf16
Latest completed XL degradation	<code>photo_v3_noise_mix</code>

6 Completed Results

6.1 Stage 1 VAE

The selected Stage 1 VAE checkpoint is `autoencoder_photo10k_b16_eval_online/checkpoints/best_eval_recon.pt`. It was selected at the 50k-step checkpoint with:

Metric	Value
eval/recon	0.01198
eval/kl	9.38684
eval/psnr	40.19

6.2 Photo100k Baseline Stages

The first photo100k scale-up completed deterministic latent pretraining, conditional diffusion, and condition-start fine-tuning.

Stage	Run	Finished step	Selected result
Stage 2	<code>latent_pretrain_photo100k.b64</code>	30k	best latent loss 0.21230
Stage 3	<code>diffusion_photo100k.b32</code>	60k	decoded PSNR 24.56 diagnostic
Stage 4	<code>diffusion_photo100k.b32_stage4.condition</code>	5k	sampled val100 PSNR 25.4072

On the sampled validation comparison, the Stage 4 condition-start checkpoint slightly improved the Stage 3 checkpoint:

$$25.4072 - 25.3745 = +0.0327 \text{ PSNR.}$$

6.3 Photo100k v2 Denoise/Sharpening Stages

The `photo_v2` degradation curriculum produced lower absolute PSNR than the earlier degradation because the task is stronger, but Stage 4 improved both the Stage 3 v2 model and bicubic baseline on the sampled validation set.

Comparison	Sampled val100 PSNR
Bicubic baseline	22.4103
Stage 3 v2 SR	22.6699
Stage 4 v2 SR	22.8426

Stage 4 v2 improved over Stage 3 v2 by +0.1727 PSNR and beat bicubic by +0.4323 PSNR on the sampled validation set, with 81 wins and 19 losses in per-image comparison. Remaining artifacts included occasional color/contrast overshoot and small cyan/green sampling artifacts.

6.4 Stage 2 XL Condition Encoder

A smaller `photo_v3_noise_mix` Stage 2 run was stopped at step 12,700 after the validation latent loss plateaued near 0.282. The XL condition encoder then completed 80k steps with a larger 500M-class full-path target. Its main checkpoints were:

Checkpoint	Step	eval/latent_loss	eval/decoded_psnr
Best latent	66k	0.27230	21.38
Best PSNR proxy	72k	0.27940	21.52
Final/latest	80k	0.27593	21.51

The three Stage 2 XL candidates were then compared on the same 100 validation images by decoding condition-only outputs through the selected VAE. This comparison used fixed samples, fixed seed, and the same `latent_pretrain_photo100k_v3_noise_xl.yaml` configuration.

Candidate	Latent MSE	Mean decoded PSNR
Best latent, step 66k	0.91770	21.3828
Best PSNR proxy, step 72k	0.97295	21.5241
Latest, step 80k	0.89609	21.5062

The 72k-step checkpoint had the best decoded condition-only PSNR, so it was selected as the condition encoder for the next XL diffusion experiment.

6.5 Stage 4 XL Edge-Loss Condition-Start

The first Stage 4 XL run was then launched from the selected 72k-step Stage 2 XL condition encoder. The diffusion U-Net uses `base_channels=128`, `channel_multipliers=[1,2,4,8]`, six residual blocks per scale, and eight attention heads. Shape-compatible tensors were partially initialized from the smaller Stage 4 v2 checkpoint. The loss combines a small noise term, latent $\hat{x}_0$, decoded Charbonnier reconstruction, Sobel edge loss, and a highpass decoded loss.

Item	Value
Config	<code>diffusion_photo100k_xl_stage4_condition_v3_edge_b16.yaml</code>
Run	<code>diffusion_photo100k_xl_stage4_condition_v3_edge_b16</code>
Finished step	5000
Selected checkpoint	step 4250, best <code>eval/decoded_mse</code>
Training eval decoded PSNR	21.9872
W&B run	https://wandb.ai/jwheo/sr-diffusion/runs/nog04fwr

The selected checkpoint was sampled on the fixed 100-image validation split with condition initialization, 32 DDIM steps, and start timestep 50:

Metric	Value
Mean bicubic PSNR	22.3599
Mean SR PSNR	23.0793
Mean PSNR delta	+0.7195

Qualitatively, the XL Stage 4 checkpoint reduces noise and color contamination relative to the degraded LR/bicubic path. It is not yet a final restoration model: outputs remain visibly softer than the ground truth on fine textures. The next low-cost ablation is to resample the same checkpoint with a lower start timestep, such as 25, to test whether less denoising preserves more detail.

7 Limitations

This is an engineering report for an active experiment, not a claim of a final published method. The large-scale completed results are photo-domain only, while anime/illustration domain conditioning remains implemented but not yet validated at the same scale. The reported sampled PSNR values are useful for internal checkpoint comparison, but they are not a full benchmark suite against external SR methods.

References

- [1] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Bjorn Ommer. High-Resolution Image Synthesis with Latent Diffusion Models. CVPR, 2022.
- [2] Chitwan Saharia et al. Image Super-Resolution via Iterative Refinement. IEEE TPAMI, 2022.